Part 5 Going further with ASP.NET Core

Parts 1 through 4 of this book touched on all the aspects of ASP.NET Core you need to learn to build an HTTP application, whether that's server-rendered applications using Razor Pages or JavaScript Object Notation (JSON) APIs using minimal APIs. In part 5 we look at four topics that build on what you've learned so far: customizing ASP.NET Core to your needs, interacting with third-party HTTP APIs, background services, and testing.

In chapter 30 we start by looking at an alternative way to bootstrap your ASP.NET Core applications, using the generic host instead of the `WebApplication` approach you've seen so far in the book. The generic host was the standard way to bootstrap apps before .NET 6 (and is the approach you'll find in previous editions of this book), so it's useful to recognize the pattern, but it also comes in handy for building non-HTTP applications, as you'll see in chapter 34.

In part 1 you learned about the middleware pipeline, and you saw how it is fundamental to all ASP.NET Core applications. In chapter 31 you'll learn how to take full advantage of the pipeline, creating branching middleware pipelines, custom middleware, and simple middleware-based endpoints. You'll also learn how to handle some complex chicken-and-egg configuration issues that often arise in real-life applications. Finally, you'll learn how to replace the built-in dependency injection container with a third-party alternative.

In chapter 32 you'll learn how to create custom components for working with Razor Pages and API controllers. You'll learn how to create custom Tag Helpers and validation attributes, and I'll introduce a new component—view components—for encapsulating logic with Razor view rendering.